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Long-term outcomes and resilience in entrepreneurship education 2024: evaluating career success and adaptability

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ABSTRACT

This study examines the ongoing search for excellence, by emphasizing the crucial role that entrepreneurship and innovation play in the face of the drastically changing educational scene in 2024. These two factors are reshaping traditional educational frameworks, resulting in the emergence of a dynamic framework that will enable people to navigate a rapidly changing global context. Today, there is a growing need to evaluate and improve the standard of innovation and entrepreneurship in educational institutions using a comprehensive approach. 2024 is a critical year, as it marks a turning point in the conventional wisdom regarding education on account of changes in the economy, technology, and societal expectations. This study presents the idea of "Educational Excellence 2024," and assesses the improvement that has taken place in the calibre of creative and entrepreneurial methods in educational administration through the use of a comprehensive lens. The study closely analyses the complex relationships between quality benchmarks, strategic methods, and the changing requirements of educators and learners by examining the varied facets of educational excellence in the context of entrepreneurship and innovation. The research adopts a comprehensive approach in order to go beyond traditional evaluations; in the process, it acknowledges the mutually beneficial relationship between academic rigor, real-world application, and the development of an entrepreneurial mindset. It assesses the need for excellence in entrepreneurial education that will prepare entrepreneurs to develop business perceptions that will help them face real-life challenges. A more thorough comprehension of the efficacy of entrepreneurship programs becomes imperative, considering the large policy backing and financial allocation towards entrepreneurship education worldwide, particularly in developing nations such as India. This study aims to assess the current level of educational excellence and make practical recommendations regarding how schools may create an atmosphere that encourages creativity and entrepreneurship. Through an extensive analysis of optimal methodologies, obstacles, and developing patterns, the research adds to the current conversation with the goal of moulding an educational environment that not only fulfils but surpasses stakeholders' anticipations in the dynamic years to come, encompassing 2024 and beyond.

IMPACT STATEMENT

A comprehensive approach is becoming increasingly necessary to assess and raise the bar for innovation and entrepreneurship in educational institutions. This study entitled Long-Term Outcomes and Resilience in Entrepreneurship Education 2024: Evaluating Career Success and Adaptability evaluates the present state of educational excellence and offers useful suggestions about how educational institutions may foster an environment that promotes innovation and entrepreneurship. The research provides an in-depth explanation of the entrepreneurial environment and emphasizes the significant influence of entrepreneurship education on company performance. This approach to educational program alignment will guarantee that entrepreneurs have the necessary knowledge and skills to meet the changing demands of the

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SUBJECTS

Business, Management and Accounting; Sustainability Education; Training & Leadership; Continuing Professional Development

entrepreneurial ecosystem. In the process, it will ensure that entrepreneurs are prepared to embrace opportunities and confront challenges, promoting economic growth and innovation.

Introduction

In the ever-developing scenario of education, the pursuit of excellence is an ongoing aim, especially in the year 2024. The synergy between entrepreneurship and innovation has emerged as a catalyst for transforming traditional educational models, offering a dynamic framework that grooms individuals for the challenges of a rapidly changing global environment. As we stand at the precipice of a new era, it is necessary to adopt a holistic approach to assess and enhance the quality of entrepreneurship and innovation within educational institutions.

The year 2024 signifies the most critically transformational period in education history as the critical triggers of economic upheaval, social pressures, and revolutionizing changes in technology started tracking the very foundations of the old paradigms of teaching and learning. In this context, our study explores the idea of "Educational Excellence 2024," which applies a comprehensive lens to evaluate and improve the standard of creative and entrepreneurial actions in educational administration.

Every nation's economic policy should place a high priority on encouraging entrepreneurial endeavours since they are essential to the preservation and growth of economic activity (Izadi & Rezaei-Moghaddam, 2017; Mukhtar et al., 2021). Most young people who have ideas for potential businesses lack the skills or knowledge necessary to turn such ideas into successful ventures (Luo et al., 2022).

The moderating effect of entrepreneurial education has been noted by a number of researchers (Gupta & Sharma, 2018; Seyoum et al., 2021; Shah et al., 2020), but this influence has not been investigated in relation to the performance of entrepreneurs and their capacity for innovation. Studies so far have also not brought out the correlation between entrepreneurial education and its effect on career progress and entrepreneurial success. They also do not show how entrepreneurial training helps entrepreneurs face setbacks in their ventures with resilience and find ways to overcome the obstacles.

In order to understand the relationship between quality standards, strategic approaches, and the changing requirements of educators and students, this study will evaluate the many facets of educational excellence in the context of entrepreneurship and innovation. The study proposes to go beyond traditional evaluations by adopting a holistic approach, and recognizing the mutually profitable equation between outstanding academic inputs, their application to real life situations, and the moulding of an entrepreneurial perception.

The study aims to assess the current level of educational quality, and offer practical solutions that institutions can use to promote an atmosphere of innovation and entrepreneurship. By conducting a thorough analysis of the most effective approaches, obstacles, and new developments in the field, the study will add to the current endeavours of creating an educational environment that will not only meet but also surpass stakeholders' expectations in the near future.

Literature review

Some studies have been conducted with respect to entrepreneurial education's effect on the entrepreneurial spirit.

Studies bringing out the effect of education on entrepreneurial spirit and innovativeness

A study conducted in Indonesia (Simamora & Sulisty, 2023) examines how entrepreneurship education affects Indonesian millennials' aspirations to start their own businesses, utilizing perceived behaviour as a mediating factor. The study, which involved 557 individuals and was conducted across six provinces in Java, discovered that entrepreneurship education has a major impact on entrepreneurial inclinations. Positive effects on these intentions were seen from both Entrepreneurship Education Learning and Inspiration Tools, with perceived behaviour acting as a partial mediating factor. This shows how crucial entrepreneurship education is for encouraging entrepreneurial intents, especially among millennials,

by helping them grasp the different facets of entrepreneurship and the practical skills required to launch a business.

Another study (Madugu et al., 2023) investigates how entrepreneurship education influences students' innovation and creativity in Nigerian university settings. The data, collected by interviewing 244 undergraduate students, was analysed through multiple regressions. The result shows that entrepreneurial training is highly related to the inventiveness and creativity of students. The findings brought out the importance of entrepreneurial education in enhancing these skills, which are necessary for development and economic growth.

Another study (Liu et al., 2020) used a multi-factor model that measured the success of entrepreneurship education. The survey was conducted on 298 college students who had undertaken entrepreneurial programs in China. There was clear evidence of the impact of entrepreneurship education because of the improved entrepreneurial capabilities, reduction of entrepreneurship barriers, and changes in students' intentions towards entrepreneurship. In this study, a 'Triangle Measurement Model for the Entrepreneurship Education Effectiveness' was proposed and its value in creating and enriching entrepreneurship education programs as well as offering a common framework for comparative research, was clearly brought out.

Lu et al. (2021) investigated the influence of entrepreneurship education on the aspirations of university students to engage in entrepreneurship and tested the mediating effect of entrepreneurial competence concerning such a relationship. Data for this study consisted of information gathered from Chinese college students via questionnaire survey under the notion of planned behaviour. This study examined how different components of entrepreneurship education, such as business plan competitions, entrepreneurial practice support, and instruction on entrepreneurial education, affected a student's entrepreneurial aspirations. The findings showed that entrepreneurial ability is an important variable in mediating the influence and that entrepreneurship education benefits the students' inclination towards entrepreneurship. The study gives valuable information about the value of nurturing students' entrepreneurial ability and the effectiveness of the university entrepreneurship education program.

Qiu et al. (2023) examined the historical development and current practice of IEE in China's and Spain's higher education systems, with a focus on the post-COVID-19 era. Various problems in IEE activities are addressed, including: curricular integration, challenges to development, support from government and higher education, and how IEE can be used to support local development. It also probes into the significance of globalization in the context of IEE. It is mainly based on interviews held with two public institution experts from China and Spain.

Studies bringing out the effect of education on employment & economic growth

Gao (2021) emphasized the demand to extend entrepreneurial education at the university level to improve career prospects. The author underlines the interdependence of education and employment demands as she elaborates on the essentials, elements, and pathways to a creative entrepreneurial learning environment. She holds up a holistic approach that involves all stakeholders-the government and businesses-first in the way of education reform, capable of allowing the students to develop entrepreneurial attitudes and competencies.

The other research by Chang and Shih (2021) focuses on the development and current state of Taiwanese entrepreneurship education. The article discusses how entrepreneurship is part of the Taiwanese higher education system and stresses its role in helping students develop an entrepreneurial spirit. The paper discusses the changing role of entrepreneurship education in Taiwan's economic and social development through the various informing tactics and instruction methods adopted by Taiwanese institutions as a means to facilitate entrepreneurship.

Mahmudin (Mahmudin, 2023) talks of preparing the next generation to face the challenges that the global economy is probably going to pose. During this time of globalization and technological advancement, he emphasizes the need for inventive entrepreneurial skills and prospects. This analysis also encompasses how entrepreneurship education impacts business knowledge, entrepreneurial skills, and business ethics while analysing the value of such programs in order to prepare young minds for the

harsh challenges of the global economy. The author is also of the opinion that an entrepreneurial curriculum needs to infuse the importance of creativity, innovation, and social responsibility.

Jin (2022) focuses on how entrepreneurship education could be integrated into college courses that include political and ideological topics, by making use of an inventive teaching strategy; the study emphasizes the significance of entrepreneurial education in improving college students' employability and reducing societal constraints. The research relates to the situation and challenges that entrepreneurship education faces today in China by highlighting an urgent need for concerted efforts to introduce interdisciplinary courses incorporating entrepreneurial education. This means their entrepreneurial spirit and skills would be of more value if entrepreneurial education is integrated in the curriculum. A questionnaire survey (QS), that was given to college students as part of the research to gauge their interest areas and perceptions of the efficacy of entrepreneurship education, revealed that most of the students believe that their progress and entrepreneurial ambitions are not greatly impacted by entrepreneurship education.

Studies using various models for assessing ways of increasing the effectiveness of entrepreneurial education

Fen Yang (2022) in his paper presents a novel method for assessing college innovation and entrepreneurship education ecosystems. He uses the Monte Carlo approach for quantitative assessment, and suggests a random matrix theory. In order to better understand and improve entrepreneurship education, the study highlights the ecological perspective. Its goal is to optimize and balance different components of the educational ecosystem.

Another study (Hao, 2021) created a model using fuzzy neural network algorithms to teach innovation and entrepreneurship. It thus sought to improve this learning model's capacity for adaptive scheduling. In order to optimize control goal functions and analyse control constraint parameters, the study employed quantitative techniques and label recognition technologies for model control.

Lu (2022) explores how to improve college students' learning about innovation and entrepreneurship by using random matrix methods and synergetic theory. He highlights a holistic strategy that includes information sharing, resource cooperation, and organizational collaboration. He discusses the growth and difficulties in entrepreneurship education, and suggests methods for collaborative learning to increase the potency of these college programs.

An investigation (Jiang et al., 2015) into the problems and solutions in the field of innovation and entrepreneurship education in the universities in Hebei Province draws attention to problems like the utilitarianism of innovation entrepreneurship contests and the absence of assessment procedures in company incubators.

Brush (2021) emphasizes how crucial it is to build outreach initiatives, integrate extracurricular and curricular activities, and create entrepreneurial ecosystems within academic institutions. She looks at Babson College's entrepreneurship teaching strategy and shows how it combines several academic fields to generate creative chances.

Gao et al. (2021) developed a conceptual model of "teaching quality–student satisfaction," to maximize the quality of instruction for the Fundamentals of Entrepreneurship courses. It contained a thorough examination of all the elements—teaching content, delivery strategies, environment, and administration—that affect students' level of satisfaction. A student satisfaction questionnaire was used in the study, and SPSS.20 software was used for data analysis. The investigation ends with recommendations for raising both the calibre of instruction in these courses and student happiness.

Another study (Wei et al., 2019) investigates how students' inventiveness is affected by entrepreneurial instruction. It specifically looks at the various ways in which political acumen and awareness of business opportunities mediate the relationship between perceived entrepreneurship education and innovation. The researchers used structural equation modelling to analyse data gathered from Chinese student entrepreneurs. Notable discoveries encompass the affirmative correlations between innovation and entrepreneurship education, as well as the intermediary functions of political acumen and recognition of entrepreneurial prospects in these associations.

Another study (Long et al., 2021) investigates how Chinese college curriculum incorporates entrepreneurial instruction into its framework for political and ideological education. It looks at how this integration can improve college students' psychological traits, inventive thinking, and entrepreneurial skills. In order to cultivate entrepreneurial spirit and capacities, which guarantee success in a competitive setting for learners, effective entrepreneurship education should focus on aspects of entrepreneurship education. The study adopted the questionnaire survey method to collect data and finds interest in three main areas: university instructors' involvement in teaching entrepreneurial education, students' interest in entrepreneurship education, and the usefulness of entrepreneurship education in ideological and political classes. According to the survey, a considerable proportion of students express little interest in entrepreneurship education and believe it has little bearing on their personal development. The study also reveals that classroom instruction is the primary mode, and there aren't enough hands-on activities. This reduces the engagement of the student and lowers their perception level relating to the efficiency of the program. Overall, the paper develops, within the framework of ideological and political education, a more practical and integrated approach to entrepreneurial education. This strategy gives great emphasis to the importance of theoretical knowledge and actual practice in entrepreneurship education with an outlook to help the college students to develop an all-round attitude as entrepreneurs.

A study by Porkodi et al., 2023 discusses how higher education fosters the entrepreneurial desire of the students by creative inclination. In the meta-analytic review of 24 studies, it draws findings about the connection between school environment and the results of entrepreneurship. Results indicate that HEIs play an important role in cultivating the imagination of students, which is an essential quality for entrepreneurship, along with passion. The study emphasizes the positive relationship between the support and results from entrepreneurship education, and bring out the importance of HEIs in providing students with an entrepreneurial mentality and skills.

Another research study (Yu, 2018) compares the ecosystems for entrepreneurship education for China and the US. The Triple Helix Model of government, business, and university ties applies in this research study to investigate how policies of each country help or impede entrepreneurship education. This study concludes that the US ecosystem allows for a balanced foundation for entrepreneurship education and is more effective because of a bottom-up approach. Government policies have, over the past years, accelerated the diffusion of entrepreneurship education through top-down means in China; however, in terms of applicability, such a robust ecosystem shall have to be developed to offset the equity of the US model.

The value of early exposure of children to sustainable entrepreneurial education is analysed by Aral & Arslan (2022). In their opinion, if the values of entrepreneurship are inculcated among children at a very young age, sustainable entrepreneurial ventures will probably be achieved in future. Their study emphasizes the possible influence of early entrepreneurship education on sustainable development and the significance of incorporating such education into early childhood curricula to promote a culture of innovation and sustainability by reviewing relevant literature and thematically organizing findings.

Research gap

Our secondary research shows the following research gaps:

1. Impact of Entrepreneurship Education in the Long Run: The studies do not investigate the long-term effects of entrepreneurship education on students' career paths and levels of success in their entrepreneurial endeavours.
2. Resilience and Adaptability – Previous studies do not examine how students' resilience and adaptability in the face of business setbacks are affected by entrepreneurship education.

Research objectives

1. To evaluate how entrepreneurship education affects graduates' long-term professional performance, including the launch and expansion of new ventures.

2. To examine how entrepreneurship education helps students become more resilient and adaptable, especially when it comes to overcoming obstacles and setbacks in business.
3. To assess how well students are being prepared for the dynamic and unpredictable character of the entrepreneurial career path by the existing entrepreneurship education programs.

Research methodology

The study involved both reading secondary sources from reputed research data bases as also primary research in the form of a questionnaire being administered to around 30 candidates. These same candidates had attended the Entrepreneurship Development Program conducted by Symbiosis Institute of Management Studies (SIMS), which was granted by the Department of Science and Technology, Government of India. Therefore, a judgmental sampling method was used in this research. Excel statistical tools were used to analyse the data

The questionnaire did not ask for the identity of the candidate. Written consent was taken from the participants for administering the questionnaire. The purpose of the research also was explained to them namely entrepreneurial education's effect on career success and resilience and adaptability of the candidates. The paper follows the institute's norm of not more than 10% Turnitin similarity and AI 0%

Findings and analysis

Entrepreneurial trends and the impact of global events

The bar chart also shows major trends in entrepreneurial start-ups after graduation. If the majority of entrepreneurs begin from 1 to 3 years, it points toward a strong preference for early business start-up, perhaps because of well-defined ideas, family backing, or university incubator initiatives. A steady increase over the years suggests that the majority prefer to have work experience, financial stability, or industry connections before starting out. A decline in numbers after a couple of years can signify that individuals who settle down in employment, are less adventurous, or seek alternative professions. Early starters benefit from swift action and the shelter of institutions, while delayed starters can gain valuable experience before setting up. Identifying these trends directs prospective entrepreneurs to make sound decisions on when to venture into business based on their preparedness and funding. To early starters who start their businesses from 0-3 years after graduation, there is a requirement to leverage university incubators, mentorship, and networking in securing initial funding through grants, scholarship, or family capital. Early launchers must stress innovation and adaptability because they will be subject to problems rooted in inexperience and restricted financial resources. People who prefer experience first, usually starting 3-7 years after graduation, will want to gain industry skills, build professional networks, and learn leadership competencies. Frugality in this period builds capital security on the day the business is founded, and perceiving market niches allows them to use knowledge as a differentiator. For those who wait until more than seven years hence to start, strategic planning, risk assessment, and financial security are involved. Late entrants can opt for part-time entrepreneurship or side ventures before fully switching and must leverage their experience to enter high-value or niche markets. No matter when they enter, all entrepreneurs must create a solid business plan, perform extensive market research, and get guidance from experienced professionals. Keeping themselves abreast of industry trends and constantly upgrading skills will make them competitive and enhance their prospects of long-term success (Figure 1).

The survey thus showed that there has been a noteworthy increase in entrepreneurship, with an astounding 50.8% of survey participants starting their enterprises in 2023. This pattern reflects a growing interest in entrepreneurship, driven by the economy's comeback, the heightened support for new businesses, and changing job patterns. Additionally, the data highlights the impact of major world events, such as the COVID-19 pandemic, which made 2020 a crucial year for entrepreneurship, with 24.6% of respondents launching their enterprises. This shows that entrepreneurs are flexible and resilient people who frequently use adversity as a catalyst for new ideas and company ventures.

It is also possible that the education or experiences gained by the respondents have cumulatively reached a point where they felt confident enough to start their businesses, or that external factors such as changes in technology, market needs, or financing options became more favorable.

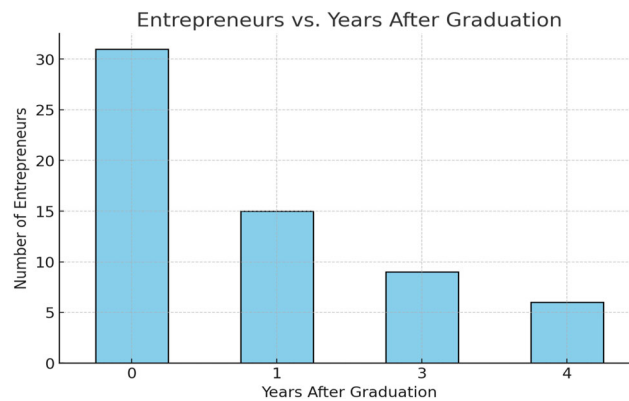


Figure 1. Time after graduation before starting own business.

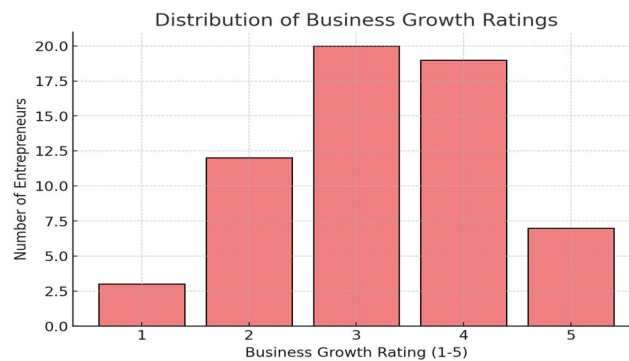


Figure 2. Business growth rate as rated by respondents.

The survey studies in detail the effects that entrepreneurship education has had on these firm owners, especially with regard to how they perceive growth, how ready they are to face obstacles, and how relevant their educational experience is overall to their entrepreneurial journey.

Growth perception and satisfaction

The bar chart shows the spread of business growth ratings provided by entrepreneurs on a scale of 1 to 5, with 1 being very low growth and 5 being very high growth. Most responses are in the moderate (3) to high (4) growth brackets, suggesting that most businesses have had steady or substantial growth. Very few entrepreneurs assessed their firm growth as very low (1), indicating that extreme slow growth is not typical, and the number of firms growing very high (5) is also small. This balanced frequency indicates a general stable entrepreneurship environment, although some firms face challenges with low growth, presumably because of issues like market competition, funding, and compliance issues. To assist low-growth firms, specific programs like financial support, mentorship initiatives, and expansion plans might be helpful. Firms whose entrepreneurs reported high or very high growth might have used successful business models, innovation, or good market positioning, and analysing their success drivers could help transfer best practices to other companies. Besides that, enhancing training programs for entrepreneurship and offering facilities for digital transformation and networking may facilitate companies in moving from moderate to high growth. More investigation into the factors influencing business growth, such as industry type, sources of investments, and operation difficulties, would yield more profound insights. Overall, the results indicate a stable but potentially improvable entrepreneurial environment, where an emphasis on eliminating barriers for low-growth firms and enhancing the success strategies of high-growth firms can lead to a more sustainable and prosperous business climate (Figures 2 and 3).

The bar graph shows how well-prepared entrepreneurs perceive their education to equip them to overcome difficulties, on a scale of 1 to 5, where 1 is poor and 5 is excellent. Most of the participants answered on a scale of 4 (Well Prepared) and 5 (Excellent Prepared), which reflects that entrepreneurship education

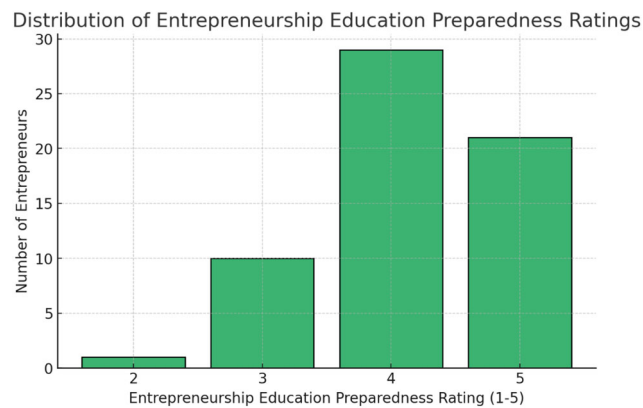


Figure 3. Attribution of business success to entrepreneurship education.

has been highly successful in making individuals capable of tackling business issues. This reflects an excellent positive image of entrepreneurial training programs. While others scored their readiness as average (3), this means that while they have been given initial knowledge, they might need to be supplemented further, e.g. with hands-on applications, mentorship, or actual world problem-solving sessions. Fewer entrepreneurs scored their readiness as unsatisfactory (1 or 2), indicating that only a minority feels thoroughly unprepared. This emphasizes the overall effectiveness of entrepreneurship education but indicates that some may find it difficult to effectively use their knowledge in business situations. To increase readiness, experiential learning avenues such as internships, case studies, and simulations should be intensified to enable entrepreneurs to cross the theory-practice divide. Participants who feel less prepared can be given special support in the form of tailored mentorship and focused workshops in business strategy, risk management, and financial planning. Additionally, learning from entrepreneurs who had self-assessed as being excellently prepared (5) would be useful in determining effective training methodologies and best practices that can be replicated to strengthen future programs. Continuous growth in curriculum planning, infusion of technology, and applicability to real life—like leadership development and digital transformation strategies—will help strengthen entrepreneurship education further. Overall, the data point to a highly effective entrepreneurial learning environment with potential areas for improvement within applicability and personalized guidance.

Facing challenges with preparedness

The split of the data reveals that nearly all entrepreneurs have faced failure or setbacks during their entrepreneurial life. The bar graph clearly reveals that the ratio of entrepreneurs responding with "Yes" to the question of whether they have faced difficulties is greatly more than responding with "No." This reflects that these difficulties in business are practically pervasive, in keeping with popular views that entrepreneur ventures are invariably beset with mishaps and difficulties. Entrepreneurial setbacks are most often learning moments for acquiring resourcefulness, adaptability, and problem-solving skills. Successful entrepreneurs attribute success to moments of failure and the subsequent capacity to learn from them and formulate improved strategies. The high rate of setbacks among entrepreneurs also highlights the need for robust support structures, including mentorship access, financial assistance, and professional training, to facilitate effective coping with business setbacks. Further research could explore sector-specific trends in an effort to determine if some sectors have more problems than others. Also, examining the relationship between challenges and long-term business success might provide valuable insights into what part setbacks play in fostering advancement. It would also be interesting to know if prior experience or business schooling helps reduce the risk or magnitude of challenges. Overall, this analysis supports that while there are problems in business, they can also be the stepping stones to enduring success (Figures 4 and 5).

The bar chart indicates how well the respondents believe their entrepreneurial education has prepared them to tackle business challenges, scaled from 1 to 5, where 1 = Poorly Prepared and 5 = Excellently Prepared. Seven in every ten participants rated their preparedness as either 4 (Well

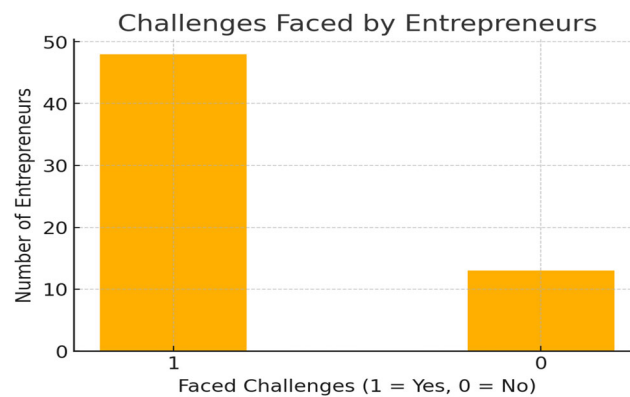


Figure 4. Respondents facing significant challenges.

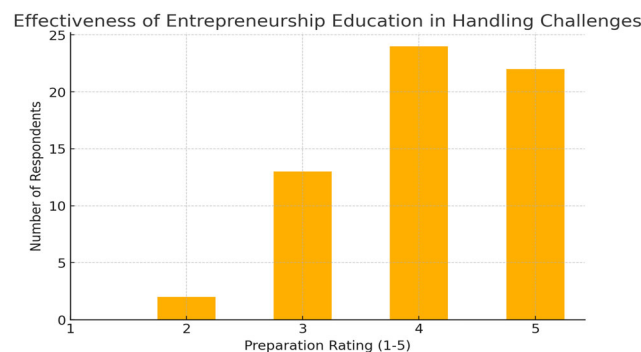


Figure 5. Ratings on handling significant challenges.

Prepared) or 5 (Excellent Prepared), which means the respondents believed their education equipped them with the capability to overcome entrepreneurial challenges. That means the curriculum provided them with practical experience, problem-solving capability, and critical thinking that were essential for a successful business. Few respondents mentioned their preparedness as 3 (Moderate Preparedness), which implies that although they partially benefited through the education, they might have experienced a gap in its applicability through practical means or usefulness in the actual world. Notably, very few or none of the participants rated themselves as 1 (Poorly Prepared) or 2 (Weak Preparedness), which highlights that most of the participants acquired at least some benefit from the education. The prevalence of high scores indicates that entrepreneurship education is performing a crucial function in establishing confidence and business knowledge among students. Still, the existence of moderate ratings presents areas where there can be improvements, like greater real-world case studies, more hands-on business experience, and incorporating mentorship programs to even better prepare would-be entrepreneurs for the future. In general, the results reaffirm that business readiness is very much facilitated by entrepreneurship education but that enhancements to experiential learning and exposure to industry may further enhance its effectiveness. A sector-by-sector or experience-based assessment may be useful in determining certain areas where the program might be improved to deliver even more value.

These findings in a way validate Porkodi et al. (2023) study wherein it was found that entrepreneurial training promoted creativity and the capacity of handling challenges in their business.

Adaptability and educational relevance

The analysis of the bar chart shows that the majority of entrepreneurs score high (4) or very high (5) in their ability to cope with changes and uncertainties, reflecting high confidence in navigating market fluctuations and business challenges. This implies that entrepreneurs possess good skills to switch strategies, navigate risks, and react well to dynamic environments. Fewer respondents scored their adaptability as moderate (3), suggesting that they do have some flexibility but can have difficulties with major uncertainties and need substantial training, guidance, or experience to further improve their flexibility.

Fewer still scored their adaptability at low (1 or 2), suggesting that there is a high level of resilience in most entrepreneurs. The findings indicate that a majority of respondents consider themselves well-equipped by their entrepreneurial education to handle the dynamism and unpredictability of the entrepreneurial environment. This reflects positively on their resilience, problem-solving skills, and overall preparedness for the challenges of entrepreneurship thanks to their entrepreneurial training. But the existence of moderate ratings indicates the urgency for additional training in crisis management, strategic planning, and risk mitigation in the industry. More adaptable entrepreneurs are more prone to maintain and grow their ventures, whereas moderately adaptable entrepreneurs may take advantage of workshops, mentorship training, and practice of real-life business situations. For additional improvement in adaptability skills, specific programs on building resilience, foresight planning, and problem-solving skills can be instituted. A more in-depth analysis by industry, level of experience, or demographics might give us a better understanding of particular areas where entrepreneurs might require more assistance (Figures 6 and 7).

The analysis of the bar chart reveals that the program of entrepreneurship education is significantly related to the professional career of most respondents, as most of them rated it 4 (Relevant) or 5 (Extremely Relevant). This reveals that the program successfully empowers individuals with entrepreneurial skills and knowledge relevant to their professional development. Yet, there were a couple of respondents who gave it 1 (Not Relevant), reflecting that for a few, the material was not relevant to their career goals. The occurrence of 3 (Neutral) implies that although a few saw the education as slightly beneficial, they did not feel it was profoundly effective. The prevalence of higher ratings attests that the program is well-designed, offering pragmatic knowledge, industry experience, and practical applications that cater to entrepreneurial and business careers. The minimal occurrence of lower ratings indicates areas of improvement, for example, more industry-focused case studies, experiential business training, and start-up incubation assistance. By streamlining the curriculum to provide more tailored learning experiences, mentorship, and exposure to cutting-edge business trends, the program can increase its applicability to a wide range of career trajectories. In general, the results suggest that entrepreneurship education is an

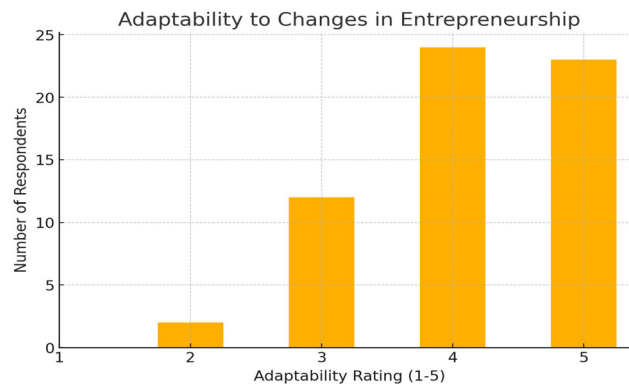


Figure 6. Adaptability rating.

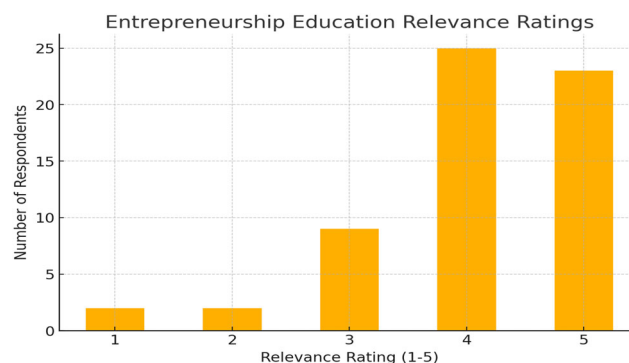


Figure 7. Relevance of entrepreneurship education.

important aspect of career development, but strategic improvements could make it even more effective. A sector-wise or demography-based comparison might yield further insight into what groups are best helped and where changes are necessary.

Connect between existing literature and study findings

Schools that embrace this entrepreneurial mindset have a significant impact on how the future is shaped. They are revamping curricula, introducing project-based learning, and promoting experiential learning that fosters creative problem-solving and critical thinking. Beyond textbooks, students can apply classroom theories to real-world situations through extracurricular activities, relationships with local businesses, and practical exposure to entrepreneurship.

Preparing future entrepreneurs is not the only aspect of this educational evolution. It involves fostering an attitude that inspires pupils to see obstacles as chances to think creatively and boldly, as well as to follow their ideas without fear. In an ever-changing culture, schools are preparing students for leadership, innovation, and positive change by encouraging an entrepreneurial attitude.

Implications for stakeholders and future directions

The survey's findings provide insightful advice to stakeholders, including legislators, business mentors, and educational institutions. The frequency of difficulties encountered by business owners emphasizes the need for strong support networks and implies that entrepreneurship education should be developed to promote resilience, adaptability, and problem-solving skills (Figure 8).

Bar chart analysis demonstrates that the program for entrepreneurship education has been greatly effective in conditioning students for the volatile and uncertain state of entrepreneurship. A clear majority of respondents evaluated the program at a rating of 5 (Very Effective), indicating that they considered it of great use in preparing them for necessary entrepreneurial capabilities. The 4 (Effective) score also garnered significant feedback, implying that participants liked the program but believed there were places for development. Interestingly, the 1 and 2 (Not Effective and Slightly Effective) ratings were either non-existent or insignificant, indicating that most participants had a good learning experience. The spread of the answers implies that the program was effective in imparting practical know-how, application-based knowledge, and useful entrepreneurial insight. The low incidence of a 3 (Neutral) rating would mean that more than half the respondents strongly believe in the program's effectiveness and are not apathetic. The high marks also confirm that the program has increased confidence for participants in exploring entrepreneurial ventures, meaning the learning is filling in the gap between theory and its implementation in real business. In spite of its success, some 4 ratings indicate some areas for improvement, including integrating more real-business case studies, simulation with hands-on experience, mentorship programs, and exposure to funding possibilities. With improvements on these aspects, the program can guarantee all participants to get the best out of it. Overall, the results show that the entrepreneurship education program is having a strong positive effect, enabling people to understand and thrive in the entrepreneurial environment. Additional sector-by-sector or demographic-based

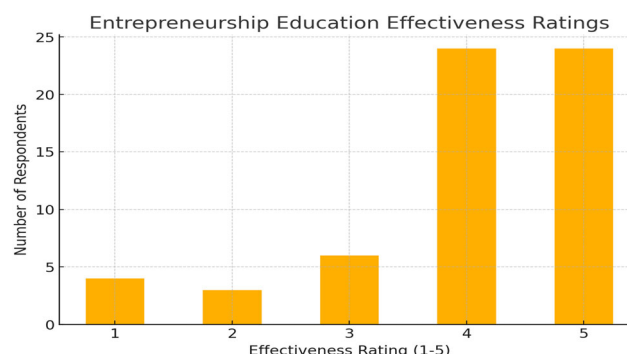


Figure 8. Effectiveness of entrepreneurship education.

analysis may offer more insight into how various groups view the program and where targeted enhancements can be achieved.

The study findings thus underscore the importance of adaptability as a key skill in the entrepreneurial toolkit, potentially guiding educators, mentors, and policymakers in focusing upon support and resources where they are most needed.

Practical applications of findings in modern educational frameworks

In line with the study findings, the following remedial steps are being taken in progressive institutions imparting entrepreneurial education:

- Curricula are being redefined in schools to foster these entrepreneurial traits. In addition to teaching information, courses are meant to foster creativity, critical thinking, and teamwork. For example, implementing project-based learning in all subject areas enables students to apply theoretical knowledge to practical situations, encouraging creativity and problem-solving abilities. By redesigning curricula to encourage innovative mindsets, schools may play a crucial role in fostering entrepreneurial thinking. This change entails going beyond conventional teaching techniques to include project-based learning, interdisciplinary learning, and real-world applications.
- Students might be encouraged to think entrepreneurially through extracurricular activities and initiatives in the classroom. These programs provide opportunities for skill development and real-world application outside of the regular classroom. Clubs, workshops, and programs that emphasize entrepreneurship provide students the chance to discover their passions, work together with classmates, and develop their leadership abilities. Students are exposed to real-world problem-solving situations through hackathons, innovation challenges, and mentorship programs, which foster resilience and creative thinking. By taking part in these activities, students can try new things, show initiative, and gain knowledge from both successes and failures in a safe setting.
- Furthermore, by allowing students to apply classroom knowledge to real-world projects and encouraging an entrepreneurial spirit that transcends academic bounds, extracurricular activities frequently close the gap between theory and practice.
- In order to foster an entrepreneurial mindset in students, it is imperative that educators receive training and assistance. In order to provide educators with the resources, approaches, and attitude required to encourage entrepreneurship in the classroom, schools should fund professional development initiatives.
- By empowering teachers to incorporate cutting-edge teaching strategies, these training programs support student-centred learning strategies that foster critical thinking and problem-solving skills. Teachers who are knowledgeable about the fundamentals of entrepreneurship are better able to coach and advise students as they pursue their entrepreneurial goals. Resources, mentorship programs, and collaborative networks are examples of support structures that help teachers better foster an entrepreneurial mindset. Teachers are able to establish stimulating learning environments where students are inspired to experiment, discover, and acquire the skills necessary for success in a world that is always changing thanks to this support.

Recommendations & the way forward

Finally, the study highlights the important impact of entrepreneurship education on business performance and offers a thorough description of the entrepreneurial ecosystem. The results thus indicate that despite the complexity and strain on the environment of entrepreneurs, education with the right background can equip such budding entrepreneurs with the resources needed to navigate in it. Educational initiatives along with the regulatory frameworks and support networks must, therefore change with the dynamic nature of entrepreneurship so as to better meet the demands of the business leaders tomorrow.

Entrepreneurial education should offer practical exposure and enhance experiential learning through internships, hands-on projects, as well as liaisons with successful entrepreneurs.

The curriculum must be up-to-date by incorporating new knowledge of course content with emerging market trends as well as technology change and the emerging principles of global business practices.

The entrepreneurial training should focus on the development of soft skills, such as communication, leadership, teamwork, as well as critical thinking skills.

Entrepreneurial education should include the practical application of theories that are followed with case studies and real projects.

The training should provide networking through contact with industry professionals, mentors, and fellow entrepreneurs.

The training should create innovative thinking through courses and workshops on innovation, creativity, and out-of-the-box ideas.

The education must necessarily include mentorship by providing access to experienced entrepreneurs and business mentors for guidance and support.

It must create an entrepreneurial mindset that will include resilience, risk-taking, and adaptability.

The training must have diverse course offerings in the form of a wide range of subjects covering various aspects of starting and running a business, from ideation to scaling.

Aligning educational programs in the above manner with the evolving needs of the entrepreneurial ecosystem will ensure that entrepreneurs are equipped with relevant skills and knowledge. This progression will guarantee that business owners are ready to take on possibilities and meet obstacles head-on, fostering innovation and economic expansion in the process.

Authors' contribution

We declare that only the three of us namely Dr. Pravin Kumar Bhoyar, Dr. Pradnya Vishwas Chitrao, and Dr. Brig. (Retd) Rajiv Divekar have conducted the study and written the research article.

We all three are responsible for the following:

- Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; AND
- Drafting the work or reviewing it critically for important intellectual content; AND
- Final approval of the version to be published; AND
- All aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

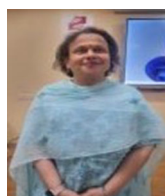
Disclosure statement

We have no interests or conflict of interests to declare

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Data availability

We agree to make data and materials supporting the results or analyses presented in our paper available upon reasonable request, unless we are unable to do so for ethical, privacy, or security concerns.

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